

INYOUNG KIM

inyoung3@illinois.edu

EDUCATION

University of Illinois at Urbana-Champaign

Aug 2024 – Present

Ph.D. in Aerospace Engineering

Advisor: Prof. Hiroyasu Tsukamoto

Seoul National University

Mar 2019 - Feb 2024

B.S. in Mechanical Engineering

B.S. in Electrical and Computer Engineering

RESEARCH INTERESTS

**Robotic autonomy and autonomous systems; nonlinear and adaptive control;
robust control under uncertainty; learning-based control with provable guarantees**

PUBLICATIONS

- [1] **Inyoung Kim**, and Hiroyasu Tsukamoto. “Closed-Loop Conformal Inference with VLM Parameter Prediction for Nonlinear Robust Adaptive Control” 2026. (Under Review)

RESEARCH EXPERIENCE

Graduate Research Assistant

2024 Aug – present

N-ACXIS Lab, Aerospace Engineering, UIUC

Advisor: Prof. Hiroyasu Tsukamoto

- Robust adaptive control of nonlinear system, implementing heterogeneous multi-drone simulation and flapper drone control

Undergraduate Research Assistant

May 2023 - Aug 2023

Radionavigation Lab, Aerospace Engineering, UT Austin

Advisor: Todd E. Humphreys

- Advanced development of aerial robot behavior

Undergraduate Thesis Research

Mar 2022 - Dec 2022

Healthcare Robotics Lab, Mechanical Engineering, Seoul National university

Advisor: Amy Kyungwon Han

- MR-compatible actuator for surgical tool manipulation

Undergraduate Research Assistant

Jul 2021 – Dec 2021

Biorobotics Lab, Mechanical Engineering, Seoul National university

Advisor: Kyu-Jin Cho

- Meter scale wave energy harvesting robot

SKILLS

Programming languages

Python, C++, Matlab

Robotics Software

ROS/ROS2, Genesis physics simulation

CAD

Solidworks

HONORS AND AWARDS

- A Department of Aerospace Engineering Stillwell fellowship** 2024
Aerospace Engineering Department, University of Illinois at Urbana-Champaign
- Awarded in recognition of research performance and academic.
- A Department of Aerospace Engineering Beatty fellowship** 2024
Aerospace Engineering Department, University of Illinois at Urbana-Champaign
- Awarded top incoming graduate students.
- Seoul National University Student Leadership Award** Feb 26, 2024
Seoul National University
- Awarded to 7 people who contributed to the university and the local community among all graduates in the first half of 2024
- The Outstanding BS thesis Presentation Award** Dec 9, 2022
Seoul National University, Department of Mechanical Engineering
- Awarded to students with outstanding B.S. thesis presentations
- Presidential Science Scholarship** Spring 2019 - Fall 2022
Sponsored by KOREA STUDENT AID FOUNDATION
- Presidential scholarship for science-talented students in Republic of Korea
- Alumni Association Scholarship** Spring 2021 - Fall 2021
Sponsored by Seoul National University, Mechanical Engineering Alumni Association
- Awarded to students who have excellent academic performance and serves as exemplary role models

LEADERSHIP, ACTIVITY AND MENTORING EXPERIENCE

- Tutor of peer tutoring program** Spring 2020 - Fall 2022, Fall 2023 (For 7 semesters)
Seoul National University
- Tutor of 'College Writing 2: Writing in Science & Technology' course
- Vice President of Department Student Council** Dec 2020 – Nov 2021
Seoul National University Mechanical Engineering
- Communicated with department Administration Office to organize department events and established better environment for students
- Undergraduate Course Assistant** Spring 2021
Seoul National University
- Assisted in solid mechanics classes by grading assignments and tests
- Run To You (Club activity for automotive)** Spring 2020 - Fall 2020
Formula/Baja Student Automotive Team from Seoul National University
- Participate 'The Korean Society of Automotive Engineers (KSAE)'s university student self-made car competition in 2020